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Key Points:

- A common pattern of snowmelt is recurrent across multiple years
- Remote sensing from multiple years can be synthesized to model the snowmelt pattern
- Principal component analysis efficiently synthesizes the snowmelt pattern

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Recurrent Snowmelt Pattern Synthesis Using Principal Component Analysis of Multiyear Remotely Sensed Snow Cover

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Abstract Snow acts as a vital source of water especially in areas where streamflow relies on snowmelt. The spatiotemporal pattern of snow cover has tremendous value for snowmelt modeling. Instantaneous snow extent can be observed by remote sensing. Cloud cover often interferes. Many complex methods exist to resolve this but often have requirements which delay the availability of the data and prohibit its use for real-time modeling. In this research, we propose a new method for spatially modeling snow cover throughout the melting season. The method ingests multiple years of MODerate Resolution Imaging Spectroradiometer snow cover data and combines it using principal component analysis to produce a spatial melt pattern model. Development and application of this model relies on the interannual recurrence of the seasonal melting pattern. This recurrence has long been accepted as fact but to our knowledge has not been utilized in remote sensing of snow. We develop and test the model in a large watershed in Wyoming using 17 years of remotely sensed snow cover images. When applied to images from 2 years that were not used in its development, the model represents snow-covered area with accuracy of 84.9–97.5% at varied snow-covered areas. The model also effectively removes cloud cover if any portion of the interface between land and snow is visible in a cloudy image. This new principal component analysis method for modeling the interannually recurring spatial melt pattern exclusively from remotely sensed images possesses its own intrinsic merit, in addition to those associated with its applications.

Plain Language Summary Mountain snow provides an important source of water. The ability to model snowmelt and the resulting streamflow helps predict the amount and timing of when water will be available for irrigation, drinking water, and other uses. Satellite remote sensing can produce maps of snow-covered area that can improve our ability to model snowmelt, but during the melt season, clouds often block the view of watersheds from space. However, snowmelt follows a repeatable spatial pattern as it melts, year after year. We used this observation to develop a model of the spatial pattern of melt using multiple years of satellite snow cover images. This model can remove the cloud interference from daily satellite snow cover images. Our model achieved 85–98% accuracy in representing the spatial pattern of snow observed in satellite images even when starting with 95% cloud cover. Other models achieve similar accuracy but require much more data to accomplish this, which prevents them from being used for cloud removal in real time. We expect the model to improve our ability to model streamflow from snowmelt runoff.

1. Introduction

Snow melt is a vital source of fresh water in many places across the world. For example, in the western United States between 39% and 67% of the annual precipitation occurs as snowfall, and between 50% and 80% of the annual flow is provided by snowmelt (Serreze et al., 1999; Stewart et al., 2004). The landscape-scale spatiotemporal pattern of snow coverage possesses tremendous intrinsic value for modeling snow processes, especially during melt, whether for real-time, retrospective, forecasting or climate change applications. The purpose of this article is to present a model which synthesized this pattern.

Satellite remote sensing of snow cover began in the 1970s to provide information for modeling. Daily imagery at a spatial resolution of 1.1 km at nadir has been available since 1979 with the Advanced Very High Resolution Radiometer (Jensen, 2016). Although this sensor had exceptional temporal resolution, the spatial resolution of 1.1 km may be too large for small basins (Rango, 1993). The launch of the MODerate Resolution Imaging Spectroradiometer (MODIS) in 1999 on board the Terra satellite initiated daily worldwide coverage at a spatial resolution of 500 m.

MODIS snow detection is hindered by two factors: intermittent unavailability and errors or uncertainty at the pixel level. These are affected by viewing conditions, land cover, the amount of snow, and cloud masking (Riggs et al., 2017). For the watershed in this research, during the annual snowmelt period lasting approximately 150 days, only 28 days on average had less than 10% cloud cover. In Austria an average cloud coverage of 63% over mountainous basins was observed during the snowmelt period (Parajka & Blöschl, 2006). Thus, cloud cover poses a substantial data availability issue. Errors at the pixel level affect accuracy, too; however, on cloud-free days the MODIS product showed 93% agreement with ground measurements (Hall & Riggs, 2007). Consequently, cloud cover is a more significant problem than pixel accuracy.

Remotely sensed imagery can be incorporated into deterministic models such as the Snowmelt Runoff Model to generate depletion curves which describe how the snow-covered area (SCA) in a watershed changes with time during a melt season (Martinec & Rango, 1986). MODIS imagery has been used to increase the accuracy of Snowmelt Runoff Model; however, cloud cover is such a persistent issue that the imagery must be processed for cloud removal or not used at all (Qiu et al., 2014; Safari Shad et al., 2014; Steele et al., 2017). Because of the importance of the spatial pattern and the significant impediment posed by clouds, much of the research on spatial representation of snow cover via remote sensing has been devoted to cloud removal. For this reason, we discuss current methods of cloud removal before presenting our spatial model. Cloud removal from MODIS images has been approached by various means including interpolation, sensor combinations, probability methods, and various combinations of these, which grow in complexity.

Interpolation methods can be divided into three main categories: spatial, temporal, and spatiotemporal methods. Spatial methods have been developed based on adjacent pixels and more complex methods. Gafurov and Bárdossy (2009) used visible pixels from a single image to interpolate values for nearby cloud-obscured pixels. Among the complex methods, the regional snow line uses visible pixels on a given day along with elevation to interpolate cloud-covered pixels (Krajčič et al., 2014; Parajka et al., 2010). This method assumes elevation to be the only influence on snowmelt, but slope and aspect play important roles as well (Hock, 2003). The regional snowline method acknowledges this by defining an elevation range considered to be partial snow (Parajka et al., 2010). Krajčič et al. (2014) found a decrease in accuracy during the month of April using the regional snow line which they attributed to patchy snow melt during the end of the melt season, as also discussed in Parajka and Blöschl (2012).

Temporal filtering methods relate a cloud-covered pixel to the same pixel from images collected at different times. The MODIS sensor is flown aboard both the Aqua and Terra Satellites. The Terra satellite has a morning overpass, and Aqua has an afternoon overpass time. Together they produce two images daily with different cloud cover percentages and locations. Cloud cover reduction by temporal interpolation between them has achieved 39.9% cloud cover removal over Europe (Dietz et al., 2012). Parajka and Blöschl (2008) found average cloud cover reductions from 63% to 52% combining Terra and Aqua. Extending the interpolation time frame using imagery from the 7 days prior to a cloudy image, Parajka and Blöschl (2008) found further cloud reduction to 4%. Gao et al. (2010) applied a temporal filter with Aqua and Terra and incorporated the Aqua Advanced Microwave Scanning Radiometer (AMSR-E) to produce cloud-free imagery. AMSR-E uses microwave observations and measures snow water equivalent, which is not hindered by cloud cover; however, it has a coarse spatial resolution of 25×25 km (Gao et al., 2010). Combinations of temporal interpolation methods and other sensor data can greatly reduce cloud cover; however, methods requiring images from dates subsequent to an image being processed prohibit real-time snowmelt modeling.

Spatiotemporal methods use a series of spatial and temporal interpolations to achieve substantial cloud reduction and in many cases produce cloud-free imagery. Gafurov and Bárdossy (2009) proposed a six-step process implementing a temporal method in steps 1 and 2 and spatial interpolation in steps 3 to 5. The final step uses a threshold day indicating over a 365-day time series whether the pixel is snow covered. A similar methodology was developed by De Ronco and De Michele (2014) which incorporates only five steps. In both of these a probability of snow is defined in the last step based on a “snow season,” the multiyear average time period of snow cover over a pixel.

Probability of snow cover has been an area of particular interest in recent years. Li et al. (2017) used a two-step method where the first step combines Terra and Aqua imagery and the second step applies a probability of snow cover based on spatial probability (a regional snowline style approach) and temporal probability. This temporal probability incorporates a weighting scheme based on prior and subsequent images where

a pixel is visible. More complex methods that incorporate image analysis methods such as the Hidden Markov Random Field to create a statistical measure to remove cloud have been developed (Huang et al., 2018). Simpler spatial and temporal combinations have also been developed but do not achieve cloud-free imagery on a daily time step (Parajka & Blöschl, 2008).

The incorporation of conditional probability of snow cover has been increasing in popularity among research groups. The idea proposed by Gafurov and Bárdossy (2009) and expanded in the development of the MODSNOW-tool, which produces daily cloud-free imagery, incorporates information from multiple water years (Gafurov et al., 2016). Monthly probabilities of snow coverage are developed based on historic MODIS imagery as well as a conditional probability which relates pixels to one another for cloud removal. The conditional probability is defined as the probability of one pixel being snow covered based on the status of other pixels in the watershed. A final step in the MODSNOW-tool uses the snow onset date and number of days until snowmelt in previous years for each pixel to identify whether it is snow covered on a given day. This method has been expanded to reject misclassified pixels and to accomplish cloud removal by applying a meteorological filter (Dong & Menzel, 2016). These are complex multistep methodologies that incorporate multiyear data. The incorporation of multiyear data is weakened by outlier snowpack years especially in the use of monthly probability and snow onset and snowmelt days. The use of interpolated meteorological data can induce errors through the interpolation method and requires a number of gauging stations in the watershed. These methods produce cloud-free imagery but may be prone to error in statistical outlier years which melt either earlier or later than the normal range; they may revert snowmelt timing toward the mean in outlier years. Inaccurate melt timing is problematic because earlier snowmelt relates to earlier discharge (Tan et al., 2011). Accurate simulation of discharge timing is critical for water management; as an example, in Idaho the timing of flow sometimes determines who has the legal right to claim an amount of water (Qualls et al., 2013). Modeling accuracy in outlier years is particularly important, yet available probability methods are likely to produce the largest errors in these years.

While the temporal conditional probability approach is effective, it overlooks one consistency in snowmelt. It has long been accepted that snow melts in the same pattern across years, independent of snow depth (Adams, 1976; König & Sturm, 1998; Luce & Tarboton, 2004; Sturm & Wagner, 2010). Although the meltout of a pixel does not always occur on the same day of year nor is the span of pixel melt-out dates fixed so as to yield a fixed duration of watershed meltout, they follow a similar distribution. That is, the spatial pattern of snow melt is consistent even though the timing may shift.

Wang and Xie (2009) observed the pattern of snowmelt along with variable snow duration and onset in the Tianshan Mountains over the period of 2001 to 2006 but did not make use of the pattern. Worth noting, Painter et al. (2009) developed the MODSCAG model which characterizes the fraction of snow cover at the subpixel level with precision of 93.7% to 100%, using the shape of the spectral signature of a combination of multiple bands (electromagnetic wavelength intervals) from MODIS. However, the shape of the spectral signature is unrelated to the spatial pattern of melt addressed here.

To the authors' knowledge, until now no research has sought to quantify the recurrent pattern of snowmelt. The research presented here distills and harnesses the spatial pattern which is embedded in multiple years of remotely sensed snow coverage, but often obscured by clouds, using principal component analysis (PCA) to derive a model which captures the snowmelt pattern. We assess the model's accuracy and illustrate one of its applications as a cloud removal tool. We believe that the derived spatial model possesses intrinsic merit on its own, beyond merit associated with specific applications.

2. Study Area and Data

2.1. Study Area

Our study area, the Upper Snake River Basin, lies in the northwestern United States (Figure 1), covers an area of 8,894 km², its elevation ranges between 1,737 and 4,194 m above mean sea level, and its mean annual runoff exceeds 4 × 10⁹ m³. These headwaters serve as an important water resource for Idaho agriculture, power generation, in-stream fish requirements, and municipal, commercial, and industrial uses. The watershed is bordered to the West by the Teton Mountains and includes the U.S. Geological Survey hydrologic units known as the Snake Headwaters, Gros Ventre, and Greys-Hobcock. Land cover over the study site

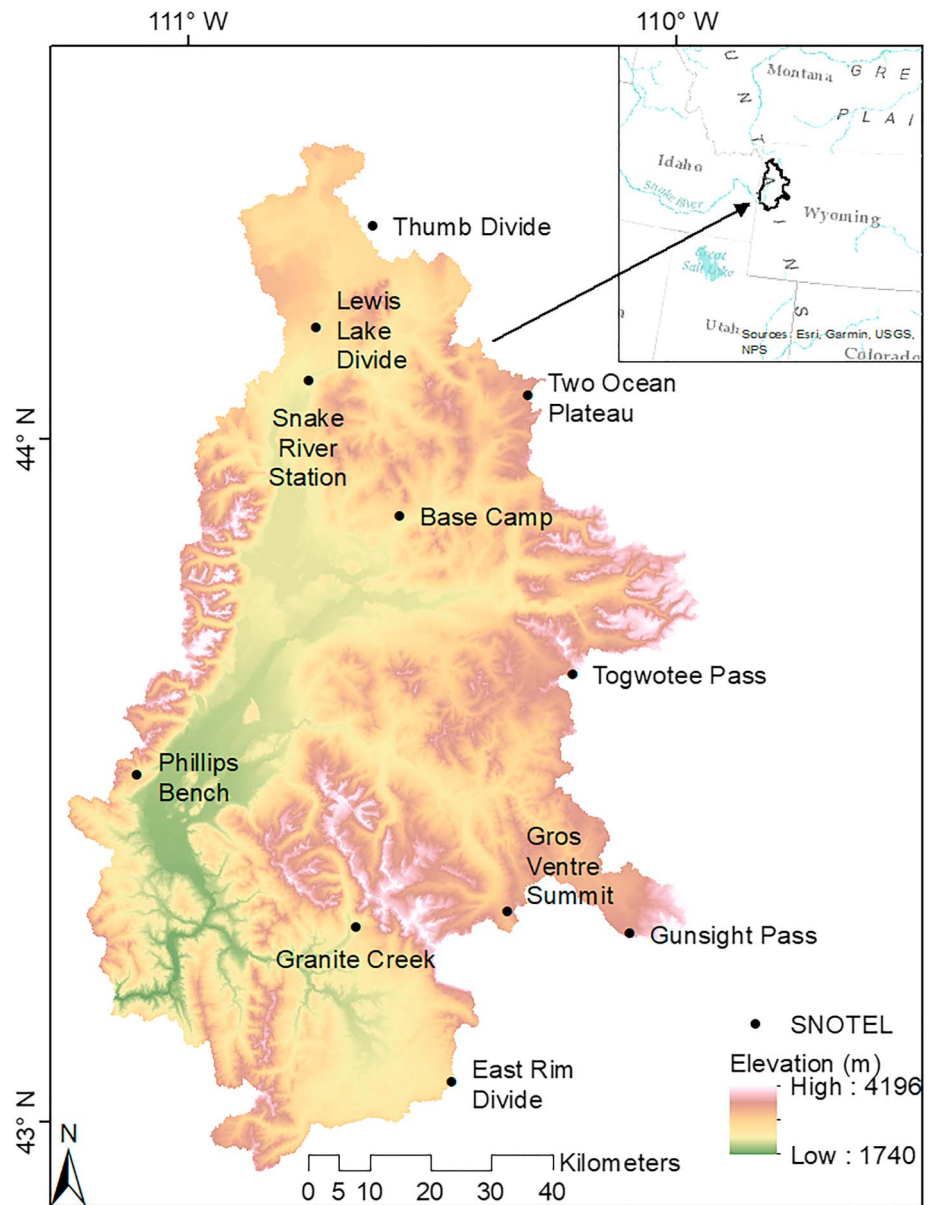


Figure 1. Digital Elevation Model of Upper Snake Basin with associated SNOTEL stations. SNOTEL = SNOW TELemetry.

is variable. Forest accounts for 47.9% of the total area. Shrub and grass land account for 45.4% of the total area with the remaining 6.7% belonging to agriculture, developed land, and wetlands. These percentages are calculated based on the 2011 National Land Cover Database (Homer et al., 2015).

Snow seasonally covers the watershed making remote sensing a useful tool for collecting snowmelt information. The snow climatology of the study area can be characterized using data collected and distributed by the Natural Resource Conservation Service from 11 ground-based SNOW TELemetry (SNOTEL) stations. Each SNOTEL site hosts a 2-m by 2-m snow pillow that measures snow water equivalent (SWE). Wide ranges of snowpack melt durations, station melt-out dates, and peak SWE values were observed across the 19 years included in this study (Table 1). Lewis Lake Divide, situated at 2,393 m above mean sea level, recorded the largest peak SWE measured within the watershed between 2000 and 2018 at 144.5 cm; its smallest recorded peak SWE was 48.5 cm. The maximum is nearly triple the minimum. Lewis Lake Divide's largest peak SWE far exceeds the maximum value recorded at Gunsight Pass of 69.3 cm, and its smallest peak SWE similarly exceeds Gunsight's smallest peak SWE of 26.7 cm even though Gunsight Pass is 600 m higher in

Table 1
Earliest and Latest Melt-Out Dates, Largest Median, and Smallest Annual Peak SWE Observed at the SNOTEL Stations at Any Time During the Study Period

SNOTEL station ID	Latest melt-out date	Earliest melt-out date	Largest peak SWE (cm)	Median peak SWE (cm)	Smallest peak SWE (cm)
Base Camp	6/10/11	5/1/01 and 5/1/07	68.8	45.0	22.1
Thumb Divide	6/16/11	4/23/15	74.9	41.9	22.6
Lewis Lake Divide	7/3/11	5/13/15	144.5	80.5	48.5
Snake River Station	6/6/11	4/29/15 and 4/29/07	58.9	44.5	23.6
Two Ocean Plateau	7/18/11	6/6/07	140.0	84.3	57.9
Togwotee Pass	7/22/11	6/6/17	108.5	60.5	48.8
Phillips Bench	7/6/11	5/19/07	111.8	61.7	47.8
Gros Ventre Summit	6/21/11	5/11/12	55.1	31.0	23.1
Gunsight Pass	7/6/11	5/18/01 & 5/18/07	69.3	34.0	26.7
Granite Creek	6/14/11	5/5/00 & 5/5/07	76.5	39.9	25.4
East Rim Divide	5/28/11	4/23/12	47.8	27.9	17.0

Note. SWE = snow water equivalent; SNOTEL = SNOW TELelemetry. Dates are formatted as MM/DD/YYYY.

elevation. It is important to note the largest peak SWE values across the 11 SNOTEL Stations were not all recorded in the same year. The largest peak SWE values, on a station-by-station basis, were recorded in 2011 (seven stations), 2017 (three stations), and 2014 (one station). The smallest peak SWE values recorded across all 19 years on a station-by-station basis were also spread across multiple years: six were recorded in 2001, five in 2010, and one in 2004. The years used for model development in this study (2000–2016) included two large snowfall years and three drought years. The day of year (DOY) on which complete meltout occurred at each station, identified as the first day of the melt season on which SWE equals zero, further documents the variability among the years (Table 1). The latest melt-out date occurred in the same year at all stations; however, the earliest melt-out dates at different stations were spread among five different years, 2000, 2001, 2007, 2012, and 2015. Large peak SWE was recorded in 2011 with a late melt, whereas 2001 saw small peak SWE and early melt. This ground-based information shows significant variability in the snowpack SWE and in melt timing and duration over the 19 years.

2.2. Data Sources

We used images from the MODIS sensor flown on the Terra satellite in this study. The version six data product, MOD10A1 (MODIS Terra snow cover daily L3 global 500-m Sin GRID V006, Hall & Riggs, 2016) distributed by the National Snow and Ice Data Center, was used. This product comes with Normalized Difference Snow Index (NDSI) snow cover, NDSI snow cover basic QA, NDSI snow cover algorithm flags QA, NDSI, and Snow albedo data. In this study only the NDSI snow cover was used. The calculation of NDSI is a combination of band 4 and band 6, which are the visible red (0.545–0.565 μm) and midinfrared wavelengths (1.628–1.652 μm), respectively (Hall & Riggs, 2015). Since no temporal combination of Terra and Aqua was applied, only the MOD10A1 product from the Terra satellite has been used in this study. Two MODIS swaths were needed to encompass the study region, h09v04 and h10v04. Fractional SCA of the total watershed was calculated as the ratio of the number of snow covered to total pixels. We used all available MODIS images from 2000 to 2018 in this study.

3. Methods

In this section, we describe the PCA model we use to synthesize the spatial melt pattern for the study watershed. We begin by explaining the data preparation methodology to distill the collection of daily images within a year into the essential information, then develop the PCA model, explain how we apply the model to an image, and finally describe how we assess the spatial accuracy of the model.

3.1. Data Preparation

In order to model the melt pattern which recurs year after year by means of PCA, we first quantified as much of the pattern as could be seen each year in the series of MODIS snow cover images given the constraints of cloud obscuration that occur within the year. We consolidated the essential pattern information from the series of daily images during a melt season by identifying 2 days of interest for each pixel. These are the last day on which snow is seen (LDS) and the first day land is seen (FDL) at each pixel during the melt season.

It is desirable to capture the pattern of first melt of the contiguous seasonal snowpack following its point of maximum spatial extent. First melt of seasonal snowpack most accurately reflects the balance of maximum SWE and cumulative melt-energy absorption at each pixel. Collectively, this balance drives the recurrent spatial pattern of melt across the watershed. Secondary snowfall events briefly reestablish snow cover after snow has melted off and represent stochastic, interannual variability; omitting them retains a larger fraction of the recurring annual melt pattern in our LDS and FDL images.

In order to identify the DOY of first melt, we began by determining a date preceding the beginning of melt, when the watershed was at or near maximum snow cover. The period of complete or maximum watershed snow cover can be identified from MODIS images within a year; however, to simplify the process, we initiated our LDS and FDL data preparation process using the day of peak SWE of a SNOTEL station (Base Camp) which typically melts out early in the season. The SNOTEL stations in this watershed all see seasonal snow cover for extended time periods, which may not be true for every watershed.

Beginning in a period of complete or maximum watershed snow cover, and considering a single pixel, we searched forward in time for the first MODIS image that reported land for that pixel and stored the MODIS image date as the FDL DOY value for that pixel. The MODIS image date from the most recent day prior to FDL on which snow was reported at that same pixel was stored as the LDS DOY value for that pixel. In actuality, we processed this search one whole image at a time. The search identified the LDS and FDL DOY for each pixel, and we stored these data into two separate output files which have the same spatial resolution and geolocation as the MODIS images. This process reduced the series of 100 to 150 daily MODIS snow cover images spanning each melt season down to two images per year, the LDS and the FDL. These images contained all the relevant information for identification of the snowmelt pattern using PCA. We prepared LDS and FDL images for each year from 2000 to 2016. The LDS and FDL raster images are similar to two images that describe the dates each pixel transitions from snow to land developed by Wang and Xie (2009) using a temporal interpolation method.

Each of these images describes the spatial pattern and temporal progression of the snowmelt. If no clouds were present during the melt season, these two images would have identical spatiotemporal patterns, except for a separation of 1 day in which each pixel went from LDS 1 day to FDL the next. Both images in this cloud-free scenario would accurately represent the melt pattern for the given year. Owing to the presence of clouds in the MODIS snow cover images, however, a pixel may be seen as snow at some point, obscured by clouds for a period, and appear as land when it is next seen. When this happens, the LDS and FDL dates for a given pixel differ by however long cloud cover persisted between LDS and FDL. This difference caused by clouds contaminates the spatiotemporal pattern exhibited in one or both of the LDS and FDL images with respect to the true melt pattern. Since it is not known whether the LDS, FDL, or both images are affected, the difference induces uncertainty in the melt pattern expressed by both the LDS and FDL images for a given year; the greater the number of days separation between LDS and FDL for a given pixel, the greater the uncertainty in that pixel's contribution to the overall melt pattern. In the data preparation stage, we intentionally ignore this uncertainty since cloud contamination is a stochastic process that behaves differently each year. In contrast, the snowmelt pattern is recurrent year after year and contributes the largest variance to the LDS and FDL images each year. By including many years, we expect PCA to isolate the annually recurring melt pattern from the stochastic cloud contamination and other noise which varies by year.

The FDL data were used to generate our final PCA model because they describe the first day on which it has been observed that a pixel has changed from snow to land. We also apply PCA to the LDS data to help verify that our PCA model primarily captures the recurrent spatial pattern and that it captures all or nearly all of it, as described later. The difference between the FDL and LDS rasters of a given year corresponds to the duration of cloud obscuration for each pixel of the corresponding year and describes the uncertainty associated with each pixel in both the LDS and FDL.

In order to describe the remainder of the process which applies to both the LDS and FDL data, we refer to LDS and FDL generically as $*D*$, unless distinguishing between them is important.

For PCA, we took each $*D*$ raster and placed it into a vector. The rows in each of these vectors corresponded to the same pixel from each year's $*D*$ raster. For example, the 35th row in one of these vectors corresponded to the same (x,y) spatial coordinate as the 35th row in every other $*D*$ vector. In the present case, there were $M = 41,503$ rows in each vector corresponding to the number of pixels in each $*D*$ image. The (x,y)

coordinates for each row were stored for subsequent reconstitution of images but were not needed in performing the PCA. The D^* vectors were concatenated into an $M \times N \times D^*$ matrix, where each column of the matrix represented one of N years, and the data across the m th row corresponded to the same (x,y) location. We included $N = 17 \times D^*$ vectors from 2000 to 2016 in forming the D^* matrix and saved the 2017 and 2018 data for verification.

3.2. PCA Snowmelt Pattern Model Development

PCA is a data reduction method commonly used in remote sensing. The D^* vectors which comprise the D^* matrix each contain a large amount of redundant or correlated information with respect to the other D^* vectors, as well as information which only appears in a few of the vectors and some information unique to individual vectors. PCA reorganizes this information into a collection of new vectors each of which is orthogonal to or uncorrelated with the others. It does this by collecting all the information correlated one way into a single vector, information correlated another way into a different orthogonal vector, and so on. Some of the vectors produced in this way may contain information which originates from a few or primarily only one of the original source vectors.

The $N \times N$ covariance matrix (or correlation matrix) of the N original D^* vectors is formed, and the N eigenvalues and eigenvectors of the covariance matrix are found. The N -year D^* matrix can be rotated into the orthogonal directions of the eigenvectors. Each of the N columns of the resulting matrix is a principal component (PC) vector produced as a linear combination of the $N \times D^*$ vectors. The PCs are uncorrelated with one another due to orthogonality. When the PC vectors are sorted in order of decreasing eigenvalues, each one contains successively less of the variance from the D^* data, in proportion to the magnitude of the eigenvalue associated with each PC. The first principal component (PC1) is aligned with the direction of maximum variance, followed by PC2 aligned with the orthogonal direction of second most variance, and so on. However, since each successive PC describes less of the overall variance, PCA usually allows one to ignore many of the later PCs while still retaining most of the variance or information present in the original data. Some of these later PCs may even contain unwanted information, such as cloud influence, or random noise, so ignoring them not only accomplishes data reduction but may also actually concentrate the results on desirable information. This method is used to condense large data sets into just a few principal components which can describe as much as 95% or more of the total variance. For an in-depth discussion of PCA refer to Everitt and Hothorn (2011).

Each D^* image describes the spatiotemporal pattern of melt that year, along with some cloud contamination. Since the range of values in the D^* corresponds to the entire duration of melt, the variance of the image primarily corresponds to the melt pattern, along with some contribution from cloud obscuration and other noise. If, as we hypothesize, the pattern of melt is similar from year to year, PCA will concentrate the melt pattern information into earlier PCs which explain a large amount of the total variance. If cloud obscuration is stochastic and varies across the years, its influence will be disbursed among different PCs. Furthermore, if cloud influence is small, the variance described by the associated PCs will also be small. The difference between FDL and LDS vectors from a given year describes the uncertainty in the values contained in that year's FDL and/or LDS vectors. With this information we can identify which PCs to keep and which ones to discard due to the influence cloud obstruction has on the confidence in a pixel's melt DOY contained in the FDL.

An important element of PCA is that it is spatially blind. PCA calculations do not rely on the spatial location of a pixel in relation to adjacent or nearby pixels. Instead, they rely entirely on the pixel's centered melt timing in relation to other colocated pixels from other years. Additionally, each D^* is purely a consolidation of observed daily image information. No spatial or temporal interpolations have been performed. PCA combines the data nonspatially and without introducing user bias since no interpolation has been performed.

Equation (1) shows the matrix multiplication used to project the $M \times N \times D^*$ matrix denoted by elements $d_{m,n}$ (far left side) into the direction of the first eigenvector $E1_n$ (second term on left) to produce the transformed values composing $PC1_m$ (right-hand vector). PC1 is aligned with the first eigenvector, and each of its rows is associated with one pixel. Equation (1) indicates the dimensions of each term, based on $N (= 17)$ years of images and $M (= 41503)$ pixels in each image. Other PCs could be calculated by substituting the other eigenvectors into equation (1). Alternatively, equation (1) could be expanded by incorporating the

additional $N - 1$ eigenvectors into the eigenvector term to form an $N \times N$ matrix. This would yield a matrix on the right in which each column is one PC.

$$\begin{bmatrix} d_{1,1} & \cdots & d_{1,N} \\ \vdots & \ddots & \vdots \\ d_{M,1} & \cdots & d_{M,N} \end{bmatrix} \times \begin{bmatrix} E1_1 \\ \vdots \\ E1_N \end{bmatrix} = \begin{bmatrix} PC1_1 \\ \vdots \\ PC1_M \end{bmatrix} \quad (1)$$

Each row of the PCs corresponds to the same pixel as the corresponding row in the $*D^*$ vectors or matrix. We could produce a raster image of any of the PCs by associating the (x,y) coordinates from the $*D^*$ vectors to the PCs. If our hypothesis is correct and a recurrent melt pattern exists across the years represented in the $*D^*$ rasters and this correlated information represents the major variance in the $*D^*$ rasters, then PCA will consolidate this pattern into one or a few of the first PCs. These PCs could be displayed as images either alone or combined with several PCs to visualize the pattern.

3.3. PCA Model Application

As noted above, we used the FDL PCs for our actual model. The values in the PCs represent “relative time of occurrence,” T^* , for each pixel of whatever phenomenon is captured by any pattern present in the PC. This is because each PC is composed of a linear combination of the FDL vectors, and each pixel in a given FDL represents a DOY. Supposing the phenomenon is melt, or conversion of a pixel from snow to land, then a pixel’s T^* represents its relative time of melt compared to other pixels in the PC; pixels with small T^* s melt earlier, and pixels with large T^* s melt later. Furthermore, pixels in a PC whose values fall within a narrow range of T^* values melt nearly concurrently. Thus, selection of an arbitrary threshold value t^* will produce a binary classification of PC model pixels as either land ($T^* \leq t^*$) or snow ($T^* > t^*$). If a spatial pattern is present in the PC representing melt, then the interface between the locus of pixels with T^* just smaller than and T^* just bigger than t^* would demarcate the PCA snowline for that t^* . For a specified t^* , each pixel can be classified as “above” the snowline ($T^* > t^*$) or “below” the snowline ($T^* < t^*$). Increasing the threshold value, t^* , classifies an increasingly greater number of pixels in the model as land and fewer as snow, and vice versa.

We exploited this threshold concept to apply the PCA snowmelt model. Application of the PCA model to a particular remotely sensed image entails finding the PC threshold value which most closely matches the snow cover pattern of the image. We match the pattern by minimizing the visible pixel error (VPE; equation (2)),

$$VPE = \sqrt{\frac{I_L^2 + I_S^2}{T_P^2}} \quad (2)$$

where T_P is the total number of visible pixels within the image and I_L and I_S are the number of pixels incorrectly classified as land and snow by the PCA model with respect to the observed image. Visible pixels are those in the image that are not obscured by cloud or missing, so that their snow/land status is observable. Since each pixel in an image corresponds to a particular pixel in the model, we can select a trial t^* value, which fixes the binary classification of each model pixel as land or snow, and compare each pixel in the model with the image to determine how many model pixels are correctly classified and how many I_S and I_L errors occur. VPE (equation (2)) can be calculated for this t^* value. We sorted the PC model vector and corresponding image pixels in order of increasing model T^* , then starting with small t^* values, increased t^* in increments which added about 1% additional watershed area in each step. At each t^* step we calculated VPE and finally selected t_m^* which minimized VPE to identify the best match PC model pattern for the observed image. For this best match, all pixels in the PC model with $T^* \leq t_m^*$ are classified as land, and those with $T^* > t_m^*$ are classified as snow. Note that t_m^* identifies the model SCA pattern for the entire spatial domain of the PC regardless of how little of the observed image was visible as expressed by T_P ; this method may be applied to cloud-free images as well as images with extensive cloud cover.

Results from cloud-obscured images are significantly better when at least some portion of the snow-land interface can be seen within the visible portion of the image. If no part of the snow-land interface is visible, the PCA model still provides some bounds on the possible location of the snowline. If some land is visible within the cloud-obscured image, but no snow, then the PCA snowline threshold t^* must lie above the

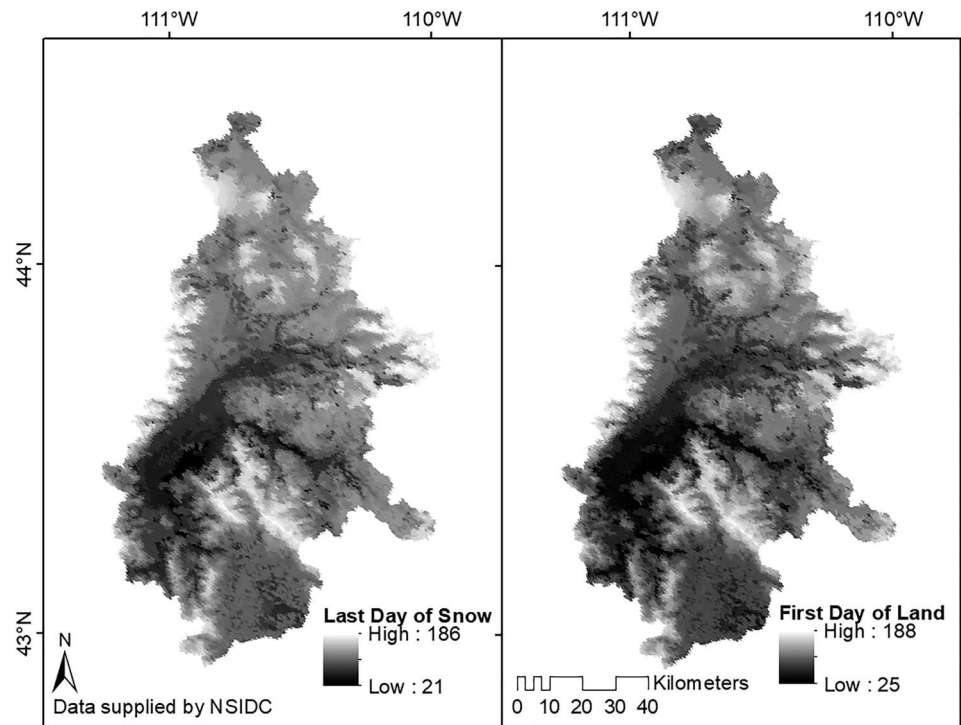


Figure 2. The 2013 images of last day of snow on left and first day of land on right. Later melt day of year values are depicted by white and earlier values by black. These images show the melt pattern which occurred during the year, along with some cloud contamination. NSIDC = National Snow and Ice Data Center.

largest model T^* associated with the visible land pixels; conversely, if snow but no land is visible in the image, then the PCA snowline threshold t^* must lie below all visible snow in the image.

3.4. PCA Model Spatial Accuracy

We calculated model spatial accuracy with respect to completely or nearly cloud-free images as a means of validating the PCA model. We also calculated model spatial accuracy for images with varying percentages of cloud cover to assess the model's cloud removal capability. For the cloud cover test, we artificially imposed cloud cover over nearly cloud-free images in order to assess the total image spatial accuracy achieved when the model was fitted to the limited number of visible pixels. This “model versus image” accuracy assessment was applied due to the limited number of ground-based sites for validation and the difference in spatial resolution between MODIS and SNOTEL (Li et al., 2017).

Once t_m^* was selected for a given image, we calculated the model spatial accuracy (A) compared to the corresponding remotely sensed image with equation (3), expressed as a percent

$$A = \left(1 - \frac{I_{Lo} + I_{So}}{T_{Po}} \right) * 100\% \quad (3)$$

We applied equation (2) to select t_m^* using only the visible pixels when an image was cloud obscured but applied equation (3) with all available pixels after the imposed cloud mask was removed to assess the total accuracy. Thus, for equation (3), the misclassified pixels I_{Lo} and I_{So} and total pixels T_{Po} were determined for the unmasked, or original image, as indicated by the subscript “o.”

4. Results and Discussion

4.1. Data Preparation

We generated LDS and FDL raster images and vectors for each year from 2000 to 2016. The images for 2013 are shown in Figure 2. Early season melt is shown as dark colors; later melt becomes progressively lighter.

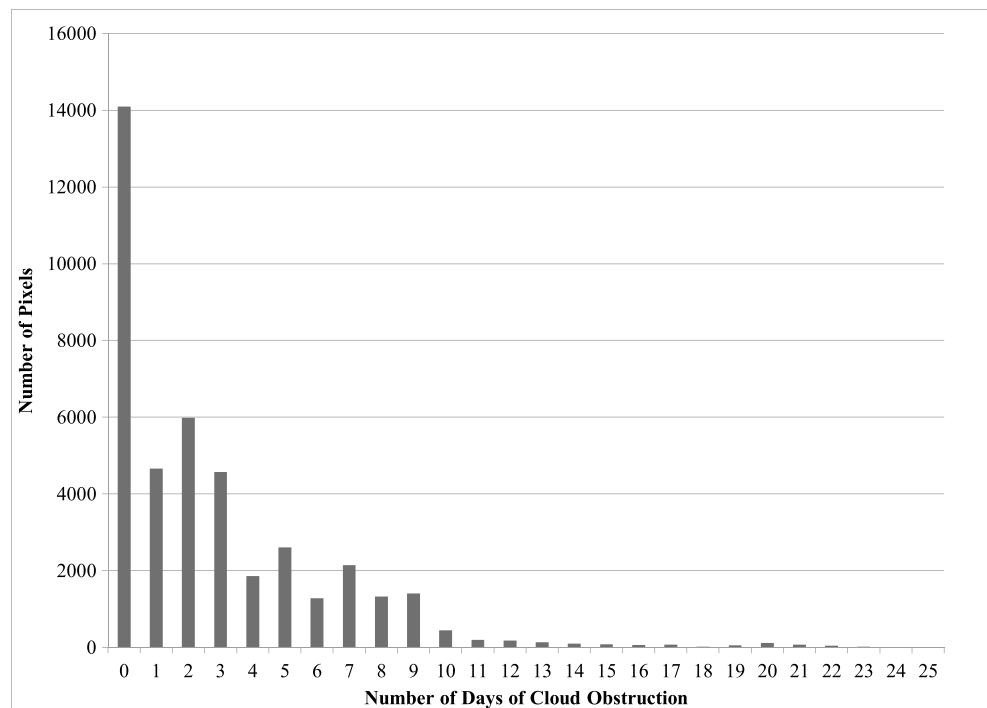


Figure 3. Frequency of number of days of cloud obscuration between first day of land and last day of snow for 2013.

Typical of all the years, these two images are similar to one another and contain the melt pattern information for a single year to the extent that it is available without interpolation or other processing to remove cloud obscuration.

The difference between the two images specifies the number of cloud-obscured days that occurred between the observed FDL and LDS for each pixel. This information is best shown in a histogram of the number of days of cloud obscuration at each pixel (Figure 3). For 2013, about one third of the pixels had no days of cloud cover separating the FDL and LDS images, meaning FDL was observed the next day after LDS. About another one third of the pixels had 1, 2, or 3 days of cloud separation, and the remainder had 4 or more days of cloud between FDL and LDS. The relative frequency of occurrence is fairly high out to 9 days of cloud separation. In Figure 2, one can also see that there are some spurious errors indicated by the small dark areas completely surrounded by light areas. This implies that the dark area melted much earlier than the surrounding pixels, sometimes as much as 100 days earlier. This phenomenon was observed every year and is discussed later.

4.2. PCA Model Development

We applied PCA based on equation (1) to process the FDL and LDS images from 2000 to 2016. PC1 is the main result of this research. We primarily used PCA results from the FDL images as our snow model, and references to PCA and specific PCs correspond to those from FDL unless otherwise noted. PC1 captured 85% of the total variance present among the 17 images for both FDL and LDS. All of the FDL vectors are similarly weighted in producing PC1. This can be seen for FDL in the PC1 weights column in Table 2. These weights are the first eigenvector components which multiply the corresponding FDL vector for each year in equation (1). In addition, PC1 is strongly correlated with the FDL vector of every year with values ranging from 0.88 to 0.94 (see PC1 correlations in Table 2). Since every FDL vector contributes similarly to PC1, the resulting PC1 is strongly correlated with every FDL vector, and PC1 explains a large amount of the overall variance among the FDL vectors; we conclude that PC1 represents a recurring pattern present in every FDL vector and that pattern represents most of the information (variance) present in the FDL images. Similar results occur for LDS.

Table 2*PCA Weights and Correlations for PC1 and PC2 Calculated From the N = 17 FDL Images Are Shown, With Variance Explained by Each PC at the Bottom of the Respective Column*

Year	PC1 weights	PC2 weights	PC1 correlation	PC2 correlation	Count of days <10% cloud	FDL – LDS average cloud duration
2000	0.225	−0.078	0.90	−0.05	2	6.66
2001	0.211	−0.105	0.88	−0.07	12	3.43
2002	0.239	0.013	0.90	0.01	7	5.34
2003	0.250	0.223	0.92	0.13	12	4.27
2004	0.310	−0.724	0.92	−0.33	13	3.23
2005	0.255	0.050	0.92	0.03	7	4.86
2006	0.199	−0.034	0.92	−0.02	12	2.63
2007	0.261	−0.220	0.93	−0.12	12	2.89
2008	0.236	0.044	0.93	0.03	11	4.68
2009	0.229	0.030	0.94	0.02	6	5.21
2010	0.286	0.244	0.93	0.12	6	7.94
2011	0.231	0.455	0.89	0.27	9	5.23
2012	0.268	−0.151	0.94	−0.08	9	2.92
2013	0.194	0.189	0.93	0.14	11	2.85
2014	0.222	0.106	0.93	0.07	11	3.04
2015	0.247	0.085	0.90	0.05	7	5.30
2016	0.229	0.101	0.94	0.06	7	3.80
Variance	85%	2%				

Note. The number of days of less than 10% cloud cover for each year during the melting season and the average number of days a pixel was cloud obscured based on the FDL – LDS differences are shown. PCA = principal component analysis; FDL = first day land; PC = principal component.

Based on their definitions, for a given year the DOY value for each pixel in the FDL and LDS images differ by the number of cloudy days separating when snow was last seen and land first appears. In other words, FDL and LDS differ based on cloud influence. We compared the PC1 vectors from both FDL and LDS and found the correlation between them to be 99.8%. They were virtually identical and described the same pattern. One of the implications of their similarity is that PCA removed all or most of the cloud contamination which differentiated the FDL and LDS images from each other leaving a common, recurrent pattern in the PC1s.

Each subsequent component is far less influential than PC1, with PC2 describing only 2% of the total variance. The variances described by PC2 through PC17 are nearly uniform and small. As shown in Table 2, PC2 has less consistency among the weights than PC1 and its correlations with FDL vectors are both small and highly variable from year to year, including some negative correlations and weights. Subsequent PCs behave similarly (not shown). This suggests that PC2 and subsequent PCs are dominated by information that is unique to one or a few FDL vectors but which contributes little to the overall variance. This would be consistent with stochastic processes such as clouds that influence each year differently and which introduce undesirable information one would prefer to filter out. It is possible that some of these higher PCs contain melt pattern information unique to one or a few years. This may be evaluated partially by examining the spatial accuracy achieved using PC1 alone to model the spatial pattern of melt of remotely sensed snow cover images, which is presented later in this article.

PC1 can be plotted using the geographic coordinates of each pixel to create a raster describing the melt pattern (Figure 4). As described in section 3.3, the numerical values in PC1 represent the *relative* time of melt, T^* , of each pixel during a melt season. Darker pixels melt earlier in the melt season, and lighter pixels melt later. A pattern is clearly visible which bears a strong topographic signature, as one would expect. A t^* contour drawn along the interface between two adjacent shades of gray would represent the snowline in the watershed at the corresponding time during a melt season. The red line in Figure 4 illustrates this for the PC1 model melt pattern producing 35% SCA. For this or any other snow-land interface line, pixels with higher numerical values (lighter shades of gray and white) would still be snow covered, and lower numerical valued pixels (darker shades of gray or black) would represent land. The isolated errors noted earlier with regard to the 2013 FDL images (Figure 2) were a recurrent feature across the years and were picked up in PC1 as can be seen in Figure 4. The black area at the northernmost tip of the watershed along with the

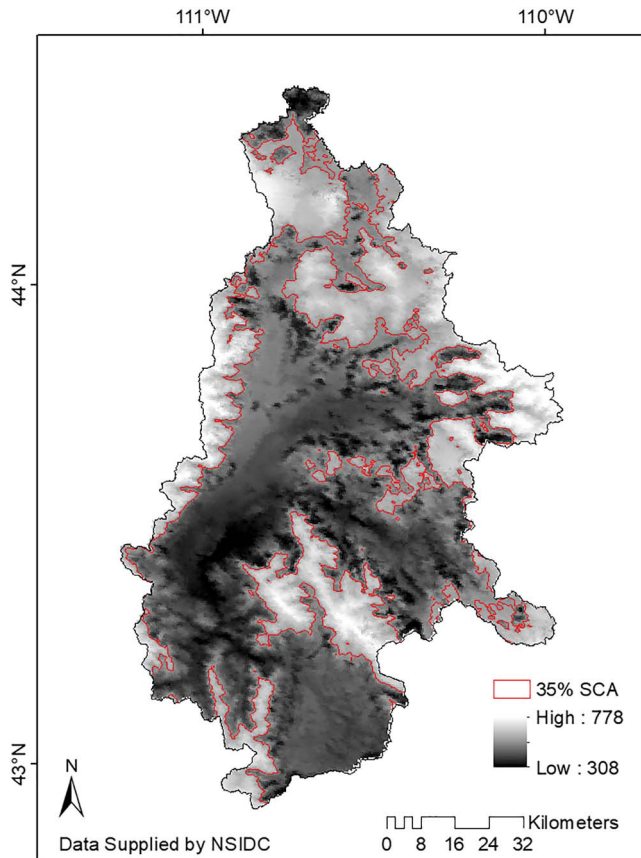


Figure 4. The first principal component image. Values represent relative time of melt, T^* . The recurrent melting pattern is seen with black (low T^* values) indicating early melt and white (high T^* values) the latest melt. The red snowline illustrates a t^* threshold separating the first principal component model snow pixels ($T^* > t^*$) from land ($T^* < t^*$), as described in section 3.3. It corresponds to $\sim 35\%$ SCA. Pixels along the snowline melt concurrently. Small dark areas surrounded by lighter pixels suggest spurious errors. SCA = snow-covered area; NSIDC = National Snow and Ice Data Center.

specs of black scattered throughout the north half of the watershed all appear to be spurious errors. Nevertheless, we believe that most errors were pushed into the second and higher PCs.

4.3. PCA Model Application and Spatial Accuracy Assessment

The PC1 spatial melt model was applied as described in section 3.3 to all remotely sensed images with less than 5% cloud cover during the melt seasons of 2017 and 2018. These years were not used in model development. To evaluate the results, the spatial accuracy achieved by the model with respect to each image was calculated as described in section 3.4. For comparison, we also applied the model and calculated the spatial accuracy of images with $< 5\%$ cloud cover from years 2000 to 2016, upon which PC1 was based. This is used to assess the relative performance of the PC1 model with independent data compared to nonindependent data. In 2017 and 2018 there were 13 and 8 days, respectively, with $< 5\%$ cloud cover during the melt season down to SCA $< 5\%$. Spatial accuracy results are presented in Table 3. In 2017 sometimes more than one image was available within a single 5% SCA increment. In these cases, averaged results were reported in Table 3 which consolidated the results into fewer than 13 entries in 2017.

In 2017, the lowest spatial accuracies ranged between about 85% and 89% for SCA between 55% and 75%. For the one image with SCA above 80%, the model spatial accuracy was nearly 92%, and for images with SCA at or below 35%, model spatial accuracy ranged between about 92% and 97%. Results from 2018 were mostly a little higher but similar to those of 2017. Results for the model applied to images from 2000 to 2016 were also similar. In those years, the average spatial accuracy was also generally lower for SCA between about 50% and 75% as they were for 2017 and 2018, with values mostly around 87% to 88%. For higher SCA, spatial accuracy ranged between about 90% and 93%; for SCA below 50%, spatial accuracies between 91% and 97% were achieved. Thus, 2017 was a little lower and 2018 was similar to 2000–2016 in the 55–75% SCA range; 2017 and 2018 were both slightly higher for SCA below 35%. The overall average accuracy for 2017 and 2018 of 91.3% and 92.5% was comparable to the

average accuracy for 2000–2016 of 91.4%. Thus, spatial accuracy of the model remained strong when applied to independent data.

Considering individual years between 2000 and 2016, the model performed similarly within corresponding general ranges of SCA (SCA $> 80\%$, $55 < \text{SCA} < 75\%$, and SCA $< 50\%$) across all years, even with interannually high variability in SWE, melt duration, and initial melt timing (results not shown).

4.4. Model Application to Cloud Removal

We applied PCA to cloud removal in the same way as done for a cloud-free day, except that the minimization of VPE (equation (2)) was carried out exclusively with the pixels visible through gaps in the clouds. Once the error-minimizing threshold t_m^* was determined, it identified the snow/land interface for the whole watershed producing a cloud-free image. In order to be able to evaluate the success of the model at cloud removal, we applied the model to the minimal cloud cover images from 2017 whose results are shown in Table 3, except with clouds artificially imposed over the images. The spatial accuracy of the model was calculated using equation (3) applied to the whole watershed.

Figure 5 provides an image demonstrating cloud removal results. The image from 31 May 2017 had 34.9% SCA shown as blue shading and 65.1% snow-free land indicated by brown shading. Sixty-six percent cloud cover was imposed over the image. The cloud-masked area is indicated by the cross hatching over the image. Equation (2) was applied solely to the cloud-free portion of the cloud-masked image.

Table 3
Total Spatial Accuracy by Snow-Covered Area Percentage and Year

SCA	Average		
	2000–2016	2017	2018
85–90	93.04		89.64
80–85	89.53	91.69	
75–80	92.02		85.06
70–75	88.37	84.90	
65–70	88.16	85.62	87.82
60–65	87.29	84.94	
55–60	90.05	89.40	
50–55	88.22		
45–50	91.63		
40–45	92.30		
35–40	89.39		
30–35	92.33	92.60	
25–30	92.05	92.14	93.79
20–25	93.38	94.49	94.12
15–20	94.04	95.19	95.82
10–15	95.97	96.17	96.59
5–10	96.68	97.13	97.52
Average	91.44	91.30	92.54

Note. Blank spaces signify that there was no data for that year (column) within the range of SCA specified (row). SCA = snow-covered area.

The model achieved 92.6% total area spatial accuracy for this image and cloud cover, and the modeled SCA was 35%, which was very close to the observed 34.9% SCA. The heavy black line traces the t_m^* threshold around the image demarcating the modeled snowline. The modeled snowline follows the observed image snowline remarkably closely, even in most areas which were masked by clouds.

In Figure 5 the most readily apparent errors occur in two places: (1) about four tenths of the way down the eastern edge of the watershed within the large protrusion, some image SCAs are enclosed within the snow-free side of the modeled snowline (I_L) and (2) just below the northern tip of the watershed, there are some snow-free image areas which lie in the modeled SCA (I_S).

Despite these localized model misclassifications, it is much more significant that the model generally traces the observed snowline closely around its entire length. There are no systematic deviations in which the observed snowline has retreated significantly above the modeled snowline on one side or end of the watershed and remained below the modeled snowline on the other, such as would appear in an instantaneous snapshot of the waterline in a bathtub as water “sloshes” back and forth, above the mean waterline on one side, and below it on the other. Such a sloshing effect might be expected if the stochastic nature of seasonal snow accumulation deposited additional snow, for example, in the southern part of the

watershed compared to the north in a particular year and then reversed or otherwise altered this stochastic pattern in other years. The PC1 model spatial pattern is fixed and would not accommodate such interannual spatial variation. We have not seen this sloshing effect between the observed images in a given year and the PC1 model. Either the interannual stochastic variation of spatially differential snow accumulation is small enough or the melt season energy budget sufficiently dominates so as to synchronize the spatial melt pattern of one side of the watershed with the other. At the scale of the study watershed, the interannual spatial melt pattern is stable and therefore the PC1 model appears to be robust.

We analyzed all of the minimal cloud cover images from 2017 by imposing percentages of cloud cover ranging from 10% to 95% in 5% increments. We were interested in assessing the capabilities of the PCA model at cloud removal in terms of its impact on spatial accuracy. We calculated the total watershed spatial accuracy when the model was applied to each observed image across the complete range of different percentages of imposed cloud. Figure 6 reports the difference between total spatial accuracy achieved by the PC1 model when applied with imposed cloud and without. The percent accuracies of the model applied to these images without imposed cloud are those reported in Table 3. As can be seen in Figure 6, spatial accuracy decreased minimally across the range of imposed cloud percentages. For most of the images, spatial accuracy decreased by less than 0.5%, and in some cases it improved slightly. Fewer than 5% of the cases experienced a decrease of spatial accuracy greater than 1%. All of these latter cases occurred for images within the lower accuracy 55–75% SCA window, except for the 84% SCA image with 65% imposed cloud whose accuracy decreased only slightly more than 1%. The only large decreases occurred at 75% imposed cloud with the 70% and 73% SCA images. In these cases, the gap in the cloud coincided with a particularly poorly performing region of the PC1 model.

A primary area of research for snow melt modeling purposes with MODIS has been cloud cover removal as discussed earlier. For this watershed in the 150-day period in which melting generally occurs, DOY 60–209, the number of days with less than 10% cloud cover is limited. A closer examination of these minimal cloud cover days reveals that they generally occur during the tail end of the melt period. Within the broader melt interval, the most critical window is when rapid melting occurs, especially between SCA of 80% and 10%. During this rapid melt period, the frequency of minimal cloud cover days is lower. This time period is most relevant for real-time snowmelt modeling because a large fraction of the total snowmelt volume flows out during this period and because this period also encompasses the peak discharge. Between 2000 and 2017, there were at least 25 days/year with less than 10% cloud cover between DOY 60 and 209. Even with this

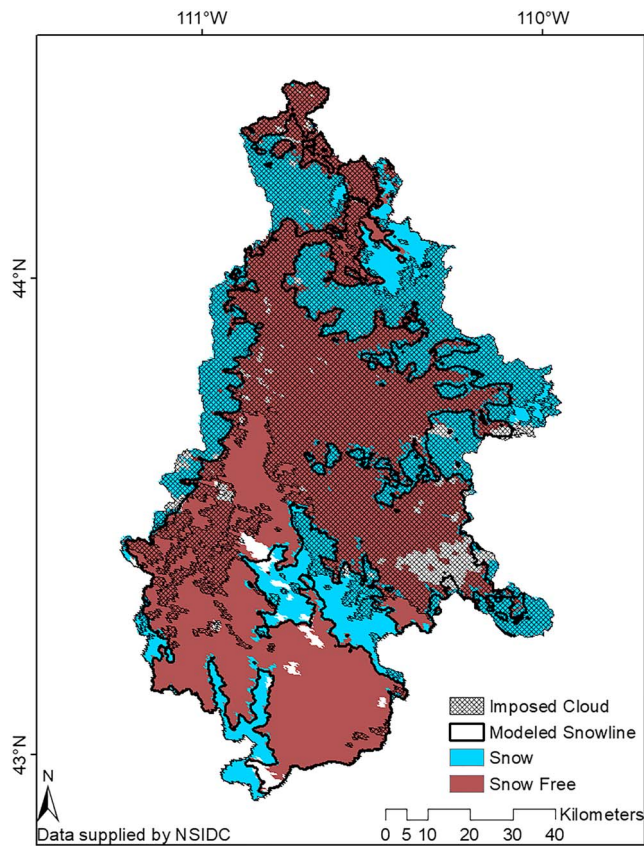


Figure 5. Observed-modeled snow-covered area (SCA) comparison. Satellite image observed 31 May 2017 has 4% cloud cover and 34.9% SCA. Imposed cloud mask covers 66% of the watershed. The first principal component modeled snowline has 35% SCA, and total watershed spatial accuracy achieved is 92.6%. White patches are actual clouds in observed image and cover 4% of the area. NSIDC = National Snow and Ice Data Center.

relatively large number of images with minimal cloud cover during the broader melt period, none of the years exceeded 13 and the average was only 9 days of minimal cloud cover during the rapid melt period (Table 2). The average duration of the rapid melt period was 66 days. This period is relatively short for the amount of change that occurs in the snowpack. The longest duration of the rapid melting period occurred in 2010 at 92 days, and the shortest occurred in 2014 at 44 days. The duration of melt does not explicitly mean small or large snowpack but rather a fast spatial melt rate once it begins. The brevity of the window of time critical to real-time modeling makes using MODIS difficult. While many methods produce cloud-free imagery, they all employ a temporal interpolation using images subsequent to the image of interest to achieve their reported accuracies. This delay to produce cloud-free imagery adds to the delay for imagery from the National Snow and Ice Data Center. While these may seem trivial, if the transition from 80% to 10% SCA is only 44 days long, a 5-day wait for images can prohibit real-time snowmelt modeling.

With our model, cloud removal can be performed as soon as an NDSI image is available. The accuracies achieved by our PCA model compare favorably to those from other research. Gafurov et al. (2016) retrieved accuracies of removing imposed cloud of 77.3–99.8% from the Karadarya Basin in the 84 days with $\leq 10\%$ cloud cover in 2008, with the lowest accuracies occurring in the transitional periods such as the melting season. These numbers were calculated under 100% cloud cover but require additional days to produce cloud-free imagery. The Adaptive Spatiotemporal Weighted Model also produces cloud-free imagery on a daily time step. This method found greater overall accuracy ranging from 93.11% to 98.92% for removing imposed cloud with a 5-day temporal filter paired with 26 meteorological stations (Li et al., 2017). Seventy-five percent of their watershed was snow-free year-round. Thus, if they recalculated accuracy exclusively for the 25% of area which experienced seasonal snow cover, their errors would be amplified and their accuracy reduced. A different study region with no meteorological stations may

cause significant reductions in their accuracy as well. A conditional probability method which incorporates interpolated meteorological data also achieved accuracies of 87–92% during the snow season using ground-based data to calculate spatial accuracy rather than MODIS imagery (Dong & Menzel, 2016).

The model presented in our research is a relatively easy technique for identifying SCA and produces agreement of 85–97.9% during the melting season. MODSCAG produces an average root-mean-square error of 5% across multiple images modeling fractional snow cover on a daily basis with added complexity (Painter et al., 2009). This varies the spectral response on a pixel-by-pixel basis to identify snow cover based on the preceding MEMSCAG model (Painter et al., 2003). Their method is useful for determining fractional SCA but requires a cloud-free day and cannot be applied for climate modeling as it requires a sensor response to produce an image.

4.5. Implications of the SCA Model

The melt pattern captured by the PCA model is truly a recurrent pattern, and our model applies independent of snowpack depth and melt timing. A criticism could be raised for application of this model had the snow pack been similar each year over the 19-year study period and then we were to apply it to more extreme years; this is not the case. Wide ranges of snowmelt duration, melt-out dates, and peak SWE values were observed at SNOTEL stations throughout the study period as described in section 2 (Table 1). Despite the significant variability in the snow climatology during the study period, the PCA model presented here performs comparably well for years with short or long melt durations, early or late melt-out dates, or high or low annual peak SWE values. In contrast, methods that use multiple years to find a probability of snow cover

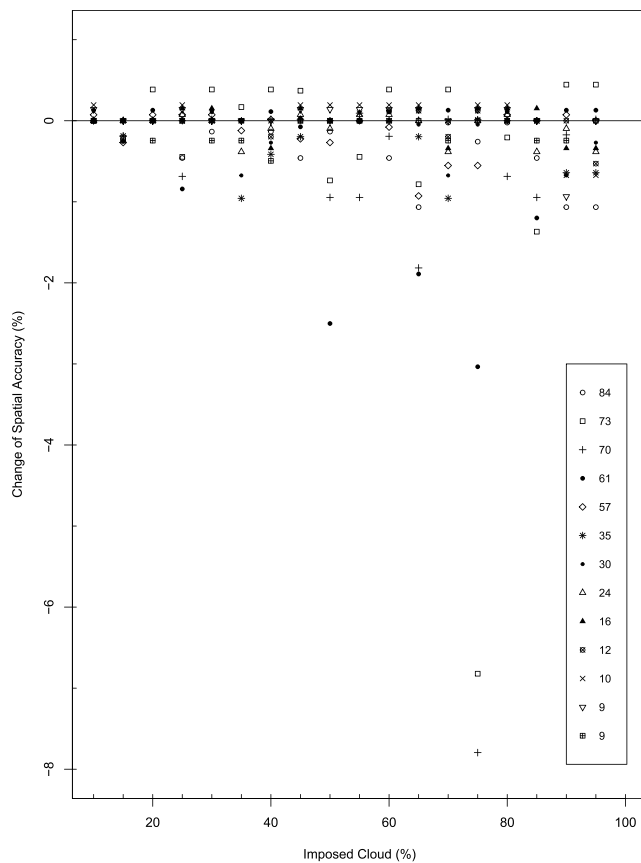


Figure 6. Change in total spatial accuracy between principal component analysis model applied to nearly cloud-free images from 2017 with imposed cloud cover and without. Actual values of total spatial accuracy without imposed cloud are those reported in Table 3 for 2017. Symbol identification corresponds to percentage of snow-covered area of the observed image.

areas, and up to 80% in other portions, still noted a pattern (Zhou et al., 2013). While the derivation of the model may change, the principal would still apply. The FDL and LDS may record earlier melt which may be due to the constant NDSI threshold used and potentially could be handled using a variable threshold for different land cover types (Painter et al., 2009; Rittger et al., 2013). In a smaller watershed a clear, repeatable pattern may be more difficult to derive due to the impact cloud cover may have. It can also be noted that the confidence in the pattern is variable throughout the melt period.

As indicated by Dong and Menzel (2016), MODIS snow products may misclassify a substantial number of pixels because of cloud/snow confusion, resulting in an overestimate of the number of snow pixels. Like the cloud pixels themselves, misclassified pixels are mostly stochastic occurrences. Consequently, the robust spatial pattern extracted by PCA may eliminate misclassified pixel contamination from the PC1 model in the same way that it removes cloud, by pushing it into PC2 or higher. However, in application misclassified pixels may still produce errors; if an image is nearly completely cloud covered and most of the visible pixels are misclassified, the PCA model may position the snowline incorrectly. On the other hand, if part of the snowline is visible, unless the misclassified pixels outnumber the correctly identified visible pixels, the PCA model will often locate the snowline appropriately and correct the misclassified pixels.

5. Conclusions

A spatially distributed melt pattern model was synthesized for a test watershed using PCA to combine multiple years (17) of remotely sensed snow cover images. PCA extracts the recurrent pattern, placing it in PC1, and pushes the influence of cloud contamination to higher PCs which are discarded. This eliminates the

based on the snow cover season add this step at the end of their cloud removal methodology since it is error prone (Gafurov et al., 2016). Paudel and Andersen (2011) used snow onset and melt-out dates to remove cloud, labeling it as a “snow cycle” step at the end of their five-step protocol. They used multiple years to group certain areas together which had similar snow cycles. Paudel and Andersen (2011) take advantage of the shift in melt timing by doing this, but the entire snow season has to pass before it can be applied. Gafurov et al. (2016) also used a multiyear approach to create a monthly probability of snow cover on a pixel by pixel basis; however, any time-dependent statistical approach may have difficulty handling a shift in melt timing. For water management, these years of low snowpack with early melt (e.g., 2001) and large snowpack with late melt (e.g., 2011) represent drought and flood years and are the most important to model accurately. Using a multiyear spatial approach rather than a multiyear temporal approach easily deals with this source of uncertainty in real time.

A significantly valuable component of our PCA method for modeling snow distribution is its minimal data requirements. The model can be developed and applied exclusively with publicly available remotely sensed images. Consequently, this model is ideal for gauged and ungauged watersheds alike. Second, the model integrates the effects of all the drivers of snowmelt on a pixel-by-pixel basis, fully representing the intraannual pattern of snowmelt, without relying on assumed proxies such as elevation. Finally, the model can be applied to an image (e.g., for cloud removal), with no other data requirements for interpolation, making it ideal for real-time modeling.

4.6. Limitations

This model was applied to a watershed that experiences complete snow coverage each year. A change in this may result in a change in the model accuracy due to different annual snow extents. However, a study conducted over an area of 950,000 km² in Eurasia encompassing Kazakhstan which receives a maximum snow coverage of 32% in some

requirement for the user to clean the input data set before analysis. Only remotely sensed images are required to derive the model, making it ideal for gauged and ungauged watersheds. The PCA model was robust when applied to independent images from 2017 to 2018, achieving accuracies from 84.9% to 97.5%. Similar accuracies were found when compared to the 2000–2016 image used to develop the model.

Cloud removal is one of many applications of the PCA model. It can be applied to images as long as some portion of the interface between snow and land is visible and produce daily cloud-free images independent of any other data source including ground-based data, meteorological data, and imagery from previous and subsequent days. Many current methods achieve high accuracies using temporal filters. These filters assume that pixels do not change from snow to land within the interpolation time frame, and to achieve their reported accuracies, they require use of images from days subsequent to the cloudy image being corrected (Gafurov et al., 2016; Li et al., 2017). This delays production of cloud-free imagery. Our model elegantly removed cloud cover based only on the visible pixels on a cloudy day because the recurrent pattern extracted by PCA is a spatially distributed description of all the drivers of snowmelt. This highlights a fundamental difference between our model and spatial/temporal filter methods. Filter methods fill cloud-obscured gaps in pixels within a remotely sensed image on a specific day; our PCA model provides a comprehensive spatial framework into which a given day's image fits, and filling its gaps is an important additional benefit.

The basis of the model is the recurrent pattern which has applications beyond cloud removal. The PCA model effortlessly handles differing snow packs, melt timing, and melt duration validating its applicability to independent and/or future years (e.g., 2017 and 2018). Applications include real-time and climate change snowmelt modeling because the model describes the spatial pattern of snow ablation throughout any melt season. In climate change simulations spatial snow cover information is often found using energy balance approaches, which could be aided by our PCA model; the model only has to be generated once and thereafter represents a consistent hydrologic process across years.

Up until now, the power of pattern recurrence across years has been largely overlooked. Many researchers have noted the existence of consistent patterns of snow melt across multiple years (Adams, 1976; König & Sturm, 1998; Luce & Tarboton, 2004; Sturm & Wagner, 2010; Wang & Xie, 2009; Zhou et al., 2013). To our knowledge, this research is the first to develop a methodology to assimilate multiyear data for the purpose of defining the recurring seasonal spatial melt pattern for a watershed. There are many applications for this model, but generation of the spatial pattern possesses its own intrinsic merit. We believe that the PCA spatial model introduces a fundamental advancement in our ability to quantify landscape-scale snowmelt processes.

Acknowledgments

All data used in this research may be obtained online through the National Snow and Ice Data Center (NSIDC) or from the Natural Resource Conservation Service SNOTEL website. The authors would like to acknowledge partial support of this research from NIFA Grant IDA01584, as well as from the Idaho Water Resources Research Institute and the DOI USGS grant number G16AP00050.

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